

National Aeronautics and
Space Administration



Earth at Night



EXPLORE
EARTH

Earth at Night

Our Planet in Brilliant Darkness

Cover image: This global view of Earth's city lights is a composite assembled from data acquired by the Suomi National Polar-orbiting Partnership (NPP) satellite. The data were acquired over nine days in April 2012 and 13 days in October 2012.
(Image credit: NASA's Earth Observatory/NOAA/DOD)

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Foreword

NASA's *Earth at Night* explores the brilliance of our planet when it is in darkness. It is a compilation of stories depicting the interactions between science and wonder, and I am pleased to share this visually stunning and captivating exploration of our home planet.

From space, our Earth looks tranquil. The blue ethereal vastness of the oceans harmoniously shares the space with verdant green land—an undercurrent of gentleness and solitude. But spending time gazing at the images presented in this book, our home planet at night instantly reveals a different reality. Beautiful, filled with glowing communities, natural wonders, and striking illumination, our world is bustling with activity and life.

Darkness is not void of illumination. It is the contrast, the area between light and dark, that is often the most illustrative. Darkness reminds me of where I came from and where I am now—from a small town in the mountains, to the unique vantage point of the Nation's capital. Darkness is where dreamers and learners of all ages peer into the universe and think of questions about themselves and their space in the cosmos. Light is where they work, where they gather, and take time together.

NASA's spacefaring satellites have compiled an unprecedented record of our Earth, and its luminescence in darkness, to captivate and spark curiosity. These missions see the contrast between dark and light through the lenses of scientific instruments. Our home planet is full of complex and dynamic cycles and processes. These soaring observers show us new ways to discern the nuances of light created by natural and human-made sources, such as auroras, wildfires, cities, phytoplankton, and volcanoes.